Functional Description

Electronic Equipment

Local Operating Panel LOP 13

Local Operating Panel LOP 14

E532404/02E

Power. Passion. Partnership.

Table of Contents

1 Safety

1.1 Important requirements for all products 4
1.2 Personnel and organizational requirements 5
1.3 Safety requirements for startup and operation 6
1.4 Standards for safety notices in the text 7

2 Functional Description

2.1 Local Operating Panel LOP – Purpose and design 8
2.2 Local Operating Panel LOP 13/14 – Function 12
2.3 Structure 13
2.3.1 Local Operating Panel LOP 13 – Controls and displays 13
2.3.2 Local Operating Panel LOP 14 – Controls and displays 17
2.3.3 Local Operating Panel LOP – Internal structure 20
2.3.4 System bus Processing Unit SPU 1 – Fuses 24
2.3.5 Service display on the front panel of Systembus Processing Unit SPU 1 25
2.3.6 System bus Processing Unit SPU 1 – Status LEDs 28
2.4 System bus Processing Unit SPU 1 – Configuration 31
2.5 System bus Processing Unit SPU 1 – Connection 35
2.6 Local Operating Panel LOP 13 – Technical data 49
2.7 Local Operating Panel LOP 14 – Technical data 50

3 Appendix A

3.1 List of abbreviations 51
3.2 Conversion tables 54
3.3 MTU contact persons/service partners 58

4 Appendix B

4.1 Index 59

DCL-ID: 0000017972 - 005

1 Safety

1.1 Important requirements for all products

Nameplate

The product is identified by a nameplate, model designation or serial number which must match the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:
- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications
- Noncompliance with the safety instructions and warning notices

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:
- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine management/parameters)
- In compliance with all safety regulations and in accordance with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use shall be considered non-intended. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Changes or modifications

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

TIM-ID: 0000040530 - 005

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.
- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

TIM-ID: 0000040531 - 003

1.3 Safety requirements for startup and operation

Safety requirements for startup

The product must be installed and accepted in accordance with manufacturers' specifications prior to initial startup.

All the requisite regulatory approvals must have been granted and all startup requirements fulfilled prior to initial startup.

Whenever the product is started, always ensure that:
- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- Wearers of cardiac pacemakers or other active medical implants are well clear.
- The operating room is adequately ventilated.
- Exhaust pipework is leak-tight and routed to atmosphere.
- Battery terminals, generator terminals and cables are guarded to preclude accidental contact.
- All personnel is clear of the danger zone represented by moving parts.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be aware of the consequences of any operations he/she performs.

During operation, the display instruments and monitoring units must be constantly observed in regard of present operating status, limit value violation and warning or alarm messages.

Malfunctions and emergency stop

Emergency procedures, in particular emergency stop, must be practiced on a regular basis.

Take the following steps if a system malfunction is detected or signaled by the system:
- Inform the duty supervisor(s).
- Evaluate the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not stay in the operating room when the product is running unless absolutely necessary and then as briefly as possible.

Keep a safe distance away from the product whenever possible. Do not touch the product unless expressly instructed to do so.

Do not inhale exhaust gases emitted by the product.

Ensure that the following requirements have been fulfilled before starting the product:
- Ear protectors are worn.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binding agent.

Operation of electrical equipment

Some components are live (high voltage) when electrical equipment is in operation.

Observe the safety instructions for these appliances.

TIM-ID: 0000040533 - 006

1.4 Standards for safety notices in the text

DANGER	In the event of immediate danger. **Consequences: Death, serious or permanent injury!** • Remedial action.
WARNING	In the event of a situation involving potential danger. **Consequences: Death, serious or permanent injury!** • Remedial action.
CAUTION	In the event of a situation involving potential danger. **Consequences: Minor or moderate injuries!** • Remedial action.
NOTICE	In the event of a situation involving potentially adverse effects on the product. **Consequences: Material damage!** • Remedial action. • Additional product information.

Safety notices

1. This manual with all safety instructions and safety notices must be issued to all personnel involved in operation, maintenance, repair or transportation.
2. The higher level warning notice is used if several hazards apply at the same time. Warnings related to personal injury shall be considered to include a warning of potential damage.

TIM-ID: 0000040578 - 004

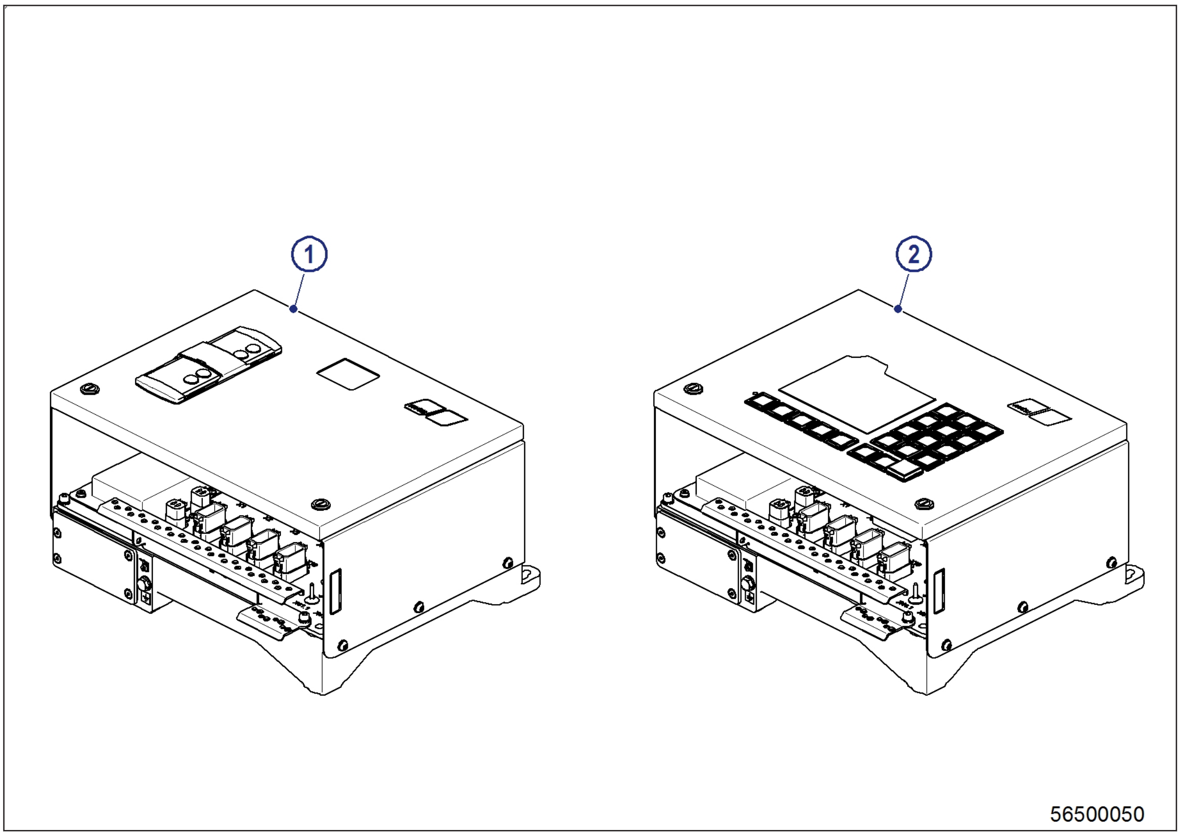
2 Functional Description

2.1 Local Operating Panel LOP – Purpose and design

Purpose

Both Local Operating Panel LOP 13 and LOP 14 serve as a central processing unit for Monitoring and Control System MCS-6 and Remote Control System RCS-6. The Local Operating Panel provides the interface to the Engine Control System ECS and acts as a local control and display unit in the engine room.

Overview

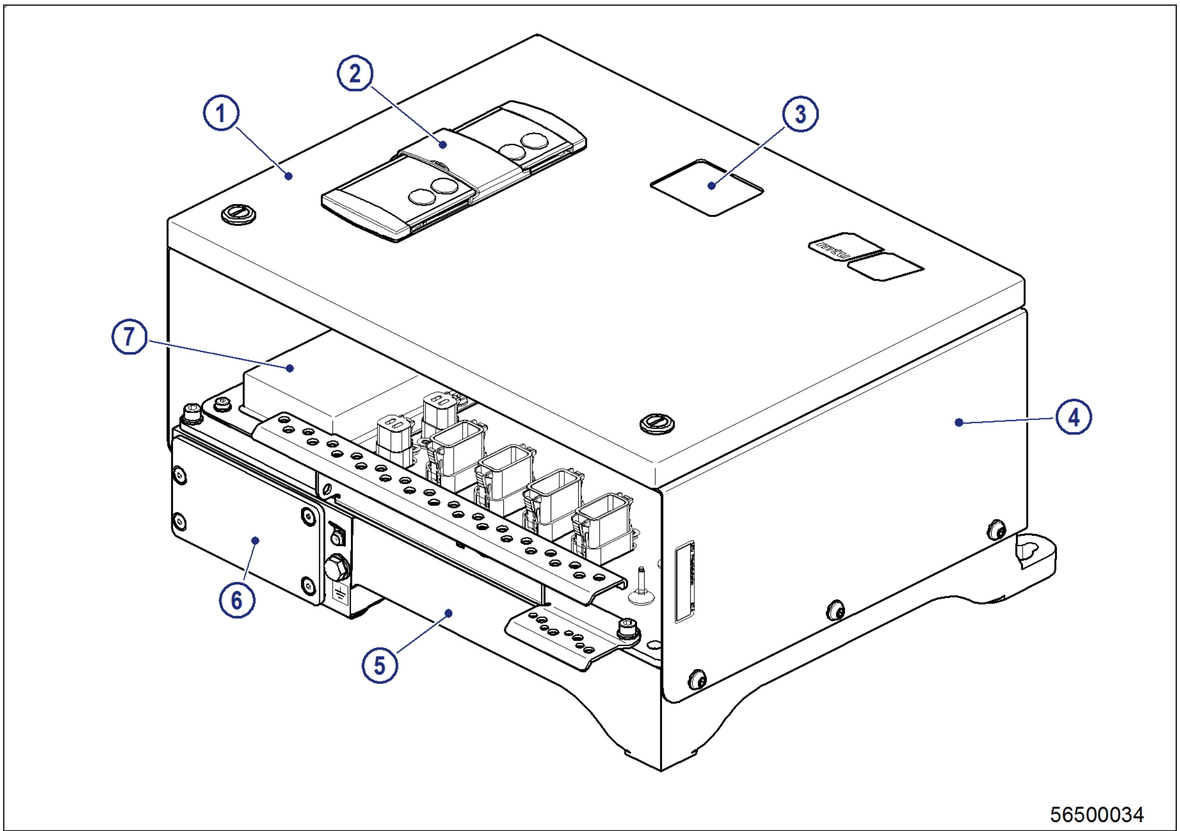


1 LOP 13 2 LOP 14

The Local Operating Panels essentially consist of:
- Front panel with control element (LOS 3 or LOS 4)
- Housing frame
- System bus Processing Unit SPU 1

TIM-ID: 0000038575 - 002

Mechanical structure of Local Operating Panel LOP 13 with SPU 1 + LOS 3

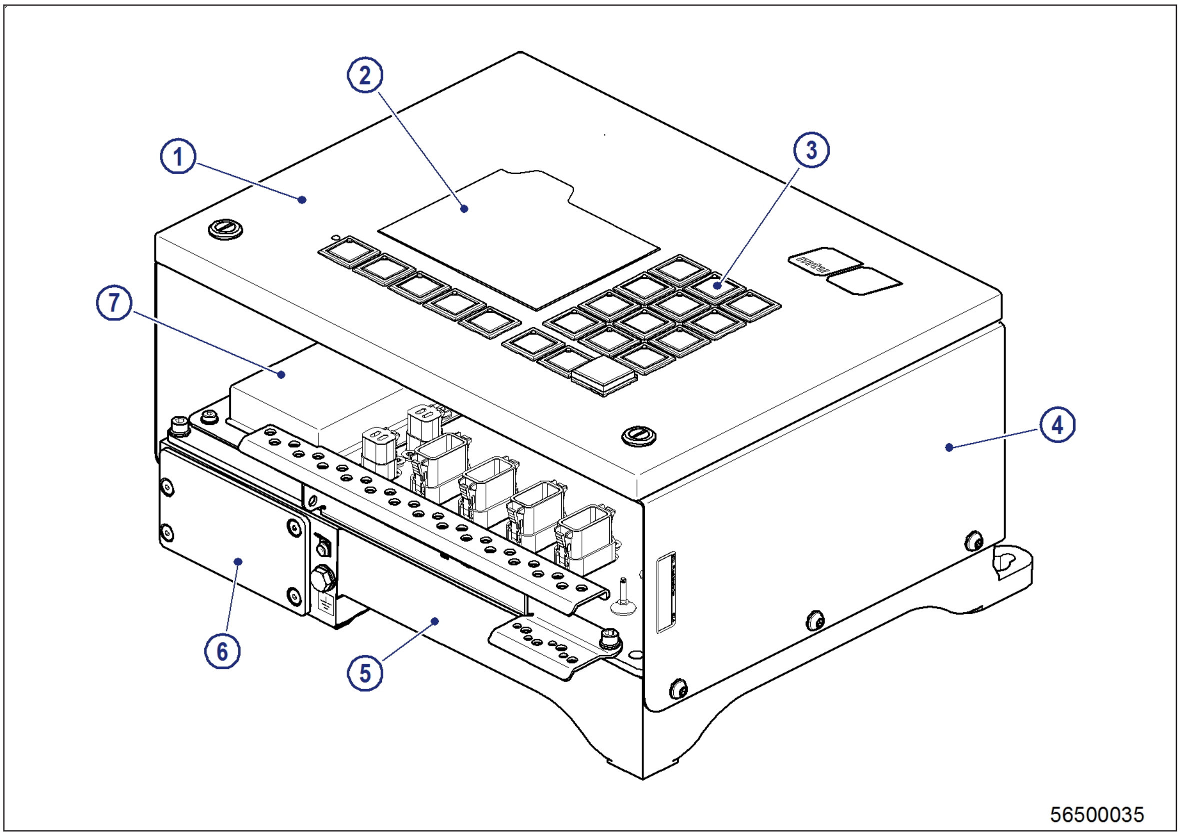


1 Front panel LOS 3-01
2 Control panel PAN
3 Service display observation window
4 Housing frame
5 System bus Processing Unit SPU 1
6 Yard slot inlet
7 Yard slot

Local control is possible using the control buttons on the operating panel of the front panel. The SPU 1 service display information can be read through the observation window of the front panel without opening the front panel.

TIM-ID: 0000038575 - 002

Mechanical structure of Local Operating Panel LOP 14 with SPU 1 + LOS 4



1 Front panel LOS 4-01
2 Display
3 Control buttons
4 Housing frame
5 System bus Processing Unit SPU 1
6 Yard slot inlet
7 Yard slot

Local control is possible using the control buttons on the operating panel of the front panel. Alarms, statuses etc. are indicated on the display.

The front panel must be opened in order to read the SPU 1 service display.

TIM-ID: 0000038575 - 002